

Collaboration and transportation resource sharing in multiple centers vehicle routing optimization with delivery and pickup

Yong Wang^{a,b,*}, Jie Zhang^a, Kevin Assogba^{c,*}, Yong Liu^{a,*}, Maozeng Xu^a, Yinhai Wang^{d,e}

^a School of Economics and Management, Chongqing Jiaotong University, Chongqing 400074, China

^b School of Management and Economics, University of Electronic Science and Technology, Chengdu 610054, China

^c School of Civil and Construction Engineering, Oregon State University, Corvallis, OR 97331, USA

^d Department of Civil and Environmental Engineering, University of Washington, Seattle, WA 98195-2700, USA

^e Transportation Data Science Research Center, College of Transportation Engineering, Tongji University, Shanghai 201804, China

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ABSTRACT

The adoption of collaboration strategies among logistics facilities and the formation of one or multiple coalitions constitute a sustainable approach to vehicle routing network optimization. This paper introduces a collaborative multiple centers vehicle routing problem with simultaneous delivery and pickup (CMCVRPSDP) to minimize operating cost and the total number of vehicles in the network. Distribution and pickup centers are allowed to share vehicles and customers in order to increase the entire network's efficiency and maximize profit. To provide the coalition coordinators with good routing solutions, we propose a hybrid heuristic algorithm which properly combines k-means and Non-dominated Sorting Genetic Algorithm-II (NSGA-II). Based on clustering solutions, the proposed Hybrid NSGA-II (HNSGA-II) first generates a real coded population to bind our mathematical model constraints and to obtain a large number of feasible solutions which converge to optimality. Chromosomes are divided for genetic operations with partial mapped crossover and swap mutation algorithms, before their recombination to ensure the quality of our results. Comparisons with the traditional NSGA-II and the Multi-Objective Particle Swarm Optimization (MOPSO) algorithm indicate better performances of HNSGA-II in terms of objective function values. We also apply Cost Gap Allocation method (CGA) and the strictly monotonic path selection principle to examine profit allocation schemes. Numerical analyses on part of Chongqing city's logistics network show the superiority of HNSGA-II over MOPSO and NSGA-II on the practical case study, as well that of CGA over the Minimum Costs-Remaining Savings (MCRS), Shapley and Game Quadratic Programming (GQP) methods. In addition, the proposed profit allocation approach has supported the establishment of a grand coalition instead of two sub-coalitions. CMCVRPSDP optimization reduces long-haul transportation, improves the vehicle loading rate and facilitates sustainable development. Through the rational allocation of profits, the proposed solution methodology assures the stability and fairness among coalition members. The implementation is also important to design sustainable urban transportation networks.

1. Introduction

Modern supply chain managers have widely considered the reduction of operation cost, the improvement of service quality and the awareness of environmental impacts as decision making objectives. Among other ways to achieve these goals, vehicle routing optimization has received significant attention and is valued as an important component of supply chain optimization. Some of the related research have focused on the optimization of distribution networks where vehicles simultaneously perform delivery and pickup activities [2,36]. The use

of the same vehicle to deliver and collect goods during the same trip significantly contributes to save cost and resource. Nevertheless, its implementation in real world environment needs improvement because simultaneous deliveries and pickups influence transportation related expenses but not other constituent of total operating cost such as the establishment, warehousing and maintenance cost. Consequently, some researchers have recently integrated collaboration among logistics facilities to minimize cost, reduce traffic congestion, increase carriers' profit and for policy control [4,29].

Cooperative game theory mainly focuses on horizontal and vertical

* Corresponding authors.

E-mail addresses: yongwx@cqjtu.edu.cn (Y. Wang), kevinassogba@gmail.com (K. Assogba), liuevery@gmail.com (Y. Liu), xmzzrxhy@cqjtu.edu.cn (M. Xu), yinhai@uw.edu (Y. Wang).

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synergies among participants [9,48]. Frisk and Gonnqvist [16] focused on forest products' transportation, encouraged collaboration between foresters and introduced a cost saving approach based on the forward and backward flows between suppliers and customers. Carriers in co-operative supply chain usually share customers and vehicles in order to optimize the use of their corresponding fleets and avoid empty truck transportation [5,12]. Therefore, the operating cost reduces and every facility makes significant benefits. Promoting collaboration between logistics facilities also aims at surviving in highly competitive environments and maximizing cost savings [26,28]. However, fairness in profit allocation and the reliability of the coordinator constitute critical issues to address for the stability of coalitions. Several approaches can separately be applied or combined to facilitate the rapid achievement of stability. Wang et al. [42] proposed the implementation of an integrated operational transportation approach consisting in combining horizontal and vertical synergies and negotiating vehicle fleets from external subcontractors in a vertical cooperation. Guajardo and Rönnqvist [19] assumed difficulties in managing grand coalitions and studied players partitioning problem in order to maintain core stability, strong equilibrium and the cardinality of each coalition.

Extensive studies were also done on Multiple Depots Vehicle Routing Problems (MDVRP) and their variants [20,38,43]. Due to difficulties engendered by the complexity of distribution networks, the search of good optimization solutions is considered as a Non-deterministic Polynomial Hard (NP Hard) problem. Therefore, the majority of research has generally designed heuristics and dynamic algorithms to increase solution quality. Kuo and Wang [25] proposed a variable neighborhood search algorithm based on stochastic initial solutions to solve MDVRP with loading cost. Nguyen et al. [32] proposed an improved tabu search algorithm with an adaptive neighborhood search strategy to assess time-dependent multi-zone multi-trip optimization solutions. Narasimha et al. [33] extended the ant colony optimization algorithm by including region partitioning to solve min-max MDVRP. Birim [6] introduced cross-docking and studied the vehicle routing problem with delivery and pickup vehicle starting and ending at cross-docks. Vehicle fleet heterogeneity was assumed, and a simulated annealing algorithm was proposed to search the best solution. Ting et al. [40] introduced the multi-vehicle selective pickup and delivery problem (MVSPDP) by relaxing vehicle constraints that require visiting every pickup node. Results suggested that solving MVSPDP yields less transportation cost than typical the pickup and delivery problem.

In order to improve service reliability, computation speed and reduce operating cost, several researchers have opted for the integration of clustering algorithm as an intermediate step in optimization and an important tool for modern logistics management [34]. Vicari and Alfó [41] used the joint clustering model of customers and products, and proposed an EM-type algorithm for ML parameter estimation. Kuo et al. [24] combined customer segmentation with heuristics to propose relevant marketing strategies. Calvet et al. [8] proposed a hybrid approach that combines statistical learning for customer demand estimation with metaheuristic algorithm, and applied their method to solve MDVRP. Defryn and Sörensen [15] proposed an improved two-level heuristic for clustered vehicle routing optimization. Assuming the number of clusters were predefined and equal to the number of vehicles, the problem mainly consisted in the design of routes in each cluster.

A comparative summary of existing research articles and the current study is presented in Table 1. Although scholars have already studied multi-depot vehicle routing problem with delivery and pickup, there are few studies which consider the formation of coalition between distribution centers (DCs) or pickup centers (PCs), and few MDVRP articles have focused on how to form more stable coalition as well as the allocation of benefit among participants. Collaboration benefits companies, but the formation process can be confronted to some challenges. Facilities may initially refuse to share their clients or

resources. Thus, the coordinator should initiate persuasive negotiation and convince participants by highlighting the importance of a coalition as well as its advantages. It is important to highlight that the rationalization of profit allocation constitutes a core subject that coordinators need to carefully approach to maintain stability.

This paper studies the vehicle routing problem before and after collaboration along with profit allocation. The introduction of collaboration between different logistics facilities extends the vehicle routing problem with simultaneous delivery and pickup (VRPSDP) to the collaborative multiple centers VRPSDP (CMCVRPSDP). Subsequently, a Hybrid Non-dominated Sorting Genetic Algorithm-II (HNSGA-II) which incorporates customer clustering is utilized to compute operating cost and number of vehicles in the existing network, and to optimize the network through collaboration and transportation resource sharing. Upon routing optimization, we search good solutions to profit allocation by applying four different methods: Shapley value model, MCRS, GQP and CGA. Finally, a practical case study of Chongqing city, China is used to evaluate the performance of our approach in real word environment.

Solving the CMCVRPSDP is a practical subject in logistics and transportation domain. Optimization solutions foster the collaboration between distribution and pickup centers and provide more systematic decisions for a better management of city logistics network. Another benefit is the reduction of transportation cost for both DCs and PCs by encouraging semitrailer truck and vehicle sharing. Besides, CMCVRPSDP optimization facilitates the reduction of traffic congestion and pollution in urban areas, and provides more profits to companies and governments.

Combining observations in Table 1, this article contributes in the following aspects: (1) We introduce a collaboration mechanism between distribution and pickup centers integrating customers, semitrailer trucks and vehicles sharing; (2) A bi-objective model to simultaneously minimize operating cost and number of vehicles is designed to evaluate the impact of collaboration on the original logistics network; (3) A hybrid non-dominated sorting genetic algorithm-II with the capability of local and global search is proposed for routing optimization, and the cost gap allocation method is used to find the best profit distribution plan for each coalition; (4) We conducted empirical analyses on an important transportation hub in China, Chongqing city, to demonstrate the use of the proposed CMCVRPSDP optimization methodology in real world.

The rest of this article is organized as follows. Section 2 illustrates the CMCVRPSDP and introduces a bi-objective model to minimize operating cost and number of vehicles. In Section 3, we design hybrid heuristic algorithm for routing optimization and introduce the cost gap allocation method for profit allocation. In Section 4, a practical study on the CMCVRPSDP in Chongqing city, China, is conducted to thoroughly analyze the importance of collaboration. The conclusion summarizes the results and proposes future research directions.

2. Problem statements and model formulation

2.1. Definition

CMCVRPSDP optimization is a tactical supply chain decision which can improve vehicle loading rate, reduce cost and generate benefit. In our study, we consider a network containing multiple DCs, PCs and customers. Fig. 1 shows the difference in the routing network before and after MCVRP optimization. The DCs and PCs are independent before collaboration, whereas they can cooperate and achieve a more organized transportation network after optimization.

As shown in Fig. 1(a), before collaboration, three DCs and two PCs are used to serve customers among which some are relatively closer to other competitors. In real life, the establishment of distribution or pickup centers is a gradual process, and as customers have already reached a significant level of confidence and loyalty towards a logistics

Table 1
Comparison between the existing literature and this study.

Study	Vehicle routing	Multiple depots/ carriers	Multiple echelons	Delivery and pickup	Collaborative logistics	Vehicle sharing	Customer clustering	Heuristic algorithm
Berger and Bierwirth [5]	✓	✓			✓	✓		
Dai and Chen [12]		✓		✓	✓	✓		
Narasimha et al. [33]	✓	✓					✓	✓
Dayarian et al. [13]	✓	✓						
Avci and Topaloglu [2]	✓			✓				✓
Birim [6]	✓	✓	✓	✓				✓
Calvet et al. [8]	✓	✓					✓	✓
Rabbani et al. [37]	✓	✓					✓	✓
Wang et al. [44]	✓	✓			✓		✓	✓
Li et al. [27]	✓	✓	✓					✓
This study	✓	✓	✓	✓	✓	✓	✓	✓

services provider, their distance to newly open and closer facilities does not influence the selection of suppliers. Moreover, logistics companies in usually develop strategies like offering discounts to attract more customers. As the strategy works, customers from different locations issue orders, and thereby increase the complexity of urban logistics networks. Besides, unreasonable customer allocation is sometimes due to the lack of durable optimization strategies to improve transportation networks fluidity. In order to achieve more sustainable networks, this paper encourages horizontal synergy among DCs and PCs.

Collaboration, as shown in Fig. 1(b), improves the fluidity and organization of the network. Proximity and vehicle capacity are evaluated to reassign each customer to the appropriate DC and PC. Pickup centers only perform pickups but distribution centers can simultaneously deliver and collect products [11]. To control the number of trips between logistics facilities, a vehicle engaged in delivery and pickup is required to collect from pickup units which were primarily assigned to the same facility and return the corresponding PC upon completion of operations. Consequently, there is no direct semitrailer truck transportation between DCs and PCs. This optimization is not only beneficial to improving distribution and collection efficiency, but also brings considerable economic benefit to each player in the coalition.

2.2. Mathematical model

In practice, third-party logistics service providers (LSP) or one of the coalition participants generally serve as coalition coordinator [44,50]. Assuming that the organization and negotiations to form coalitions are coordinated by LSP, we consider a logistics network where N distribution centers ($n \in N$) and P pickup centers ($p \in P$) respectively provide delivery and pickup services to I distribution units ($i \in I$) and J

pickup units ($j \in J$). Let D designates the set of all customer units ($D: I \cup J$). Let I' denotes the set of customer units served by vehicles that only perform delivery services, J' denotes the set of customer units served by vehicles that only perform pick-up services, and IJ denotes the set of customer units served by vehicles that perform both delivery and pickup services. A homogeneous fleet of V vehicles ($v \in V$) is deployed by each center to serve allocated units, but in case of collaboration, product flows among DCs or PCs would be assured by a set of T trucks ($t \in T$). For the formulation of a mixed integer programming model, the following parameters are introduced.

B_{ni}	=	Distance from DC n to its assigned unit i
B_{jnv}	=	Distance directly from unit j to DC n served by vehicle v
B_{pj}	=	Distance from PC p to its assigned unit j
$B_{jp v}$	=	Distance directly from unit j to PC p served by vehicle v
B_{ijv}	=	Distance from customer unit i to j served by vehicle v
B_{no}	=	Distance between DC n and DC o
B_{pu}	=	Distance between PC p and PC u
R_i	=	Demand of delivery unit i
R_i^*	=	Remaining amount of products to deliver to node i
R_j	=	Pickup order from pickup unit j
R_j^*	=	Remaining amount of products to pick from node j
S_{no}	=	Product flow between DC n and DC o
S_{pu}	=	Product flow between PC p and PC u
Q_n	=	Maximum delivery capacity of DC n
Q_p	=	Maximum pick-up capacity of PC p

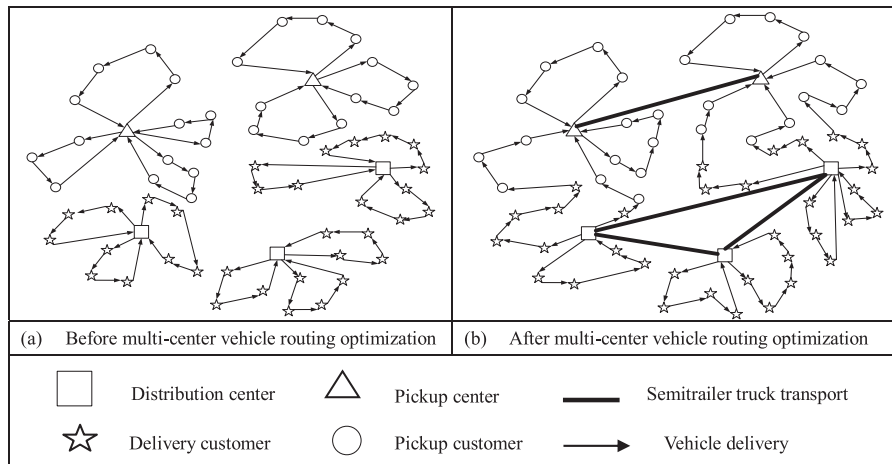


Fig. 1. Multi-center vehicle routing network optimization diagram.

Q_t	=	Maximum loading capacity on a route between DC n and DC o
Q_v	=	Maximum loading capacity of vehicles v
F_t	=	Fuel consumption rate of truck t per 100 miles
F_v	=	Fuel consumption rate of vehicle v per 100 miles
τ	=	Gas-oil price
C_t	=	Average maintenance cost of truck t
C_v	=	Average maintenance cost of vehicle v
Y_n	=	Fixed cost of DC n
Y_p	=	Fixed cost of PC p
θ	=	Variable cost coefficient
γ_h	=	Incentive provided by LSP to distribution and pickup units if collaboration occurs
Z_{not}	=	Equal to 1 if DC n and DC o cooperate, 0 otherwise
Z_{put}	=	Equal to 1 if PC p and PC u cooperate, 0 otherwise
Z_{np}	=	Equal to 1 if DC n and PC p form a coalition, 0 otherwise
x_{niv}	=	Equal to 1 if vehicle v only performs delivery services, 0 otherwise
x_{piv}	=	Equal to 1 if vehicle v only performs pick-up services, 0 otherwise
x_{ijv}	=	Equal to 1 if vehicle v only performs both deliveries and pickups, 0 otherwise

The investigated CMCVRPSDP is a practical vehicle routing problem which combines several types of integer. In addition to real integers for parameters like distances, payloads and costs, decisions on routes choices as well as collaboration are made on the basis of binary integers. Besides, in order to respect real word operations, the proposed problem is bound to consider possible incentives provided by local governments or coordinators once logistics facilities agree to collaborate. Variations and stochastic aspects of logistics operations are not assessed in this research. However, to strengthen the practicability of CMCVRPSDP, the following assumptions are included. (a) All delivery vehicles are required to leave distribution centers but vehicles engaged in pickup activities may not originate from pickup centers. Therefore, if collaboration occurs between DCs and PCs, vehicles can be used simultaneously as long as capacity constraints are not violated. (b) A vehicle departing from a distribution center cannot collect products from pickup units initially assigned to different pickup centers. In other words, even if one of the DC collaborates with two PCs and is reassigned units from them both, one vehicle should only collect from one PC's customers units.

With regards to the diversity of variables, vehicle routing problems are usually formulated using mixed-integer programming [23,47]. In accordance with the modelling approach in literature and considering the aforementioned notations, the CMCVRPSDP is designed as a bi-objective mixed-integer model which minimizes operating cost and number of vehicles. The total operating cost is sub-divided into four parts.

TC1:	Total transportation and maintenance cost for vehicles which perform pickups and deliveries
TC2:	Total transportation and maintenance cost for vehicles which perform pickups or deliveries
TC3:	Total transportation and maintenance cost for routes between DCs or PCs of the same coalition
AC:	Aggregate fix and variable cost for DCs and PCs

$$TC1 = \sum_{\substack{n \in N \\ p \in P}} \max \left\{ \sum_{v \in V} \sum_{i \in I} \frac{R_i}{Q_v}, \sum_{v \in V} \sum_{j \in J} \frac{R_j}{Q_v} \right\} \cdot \left[F_v \cdot \tau \cdot \sum_{i,j \in I} (B_{ni} + B_{ijv} + B_{jp}) \cdot x_{ijv} + C_v \cdot x_{ijv} \right]$$

$$TC2 = \sum_{n \in N} \sum_{v \in V} \sum_{i,j \in I'} \frac{R_i}{Q_v} \cdot x_{niv} \cdot (F_v \cdot (B_{ni} + B_{ijv} + B_{jnv}) \cdot \tau + C_v) + \sum_{p \in P} \sum_{v \in V} \sum_{i,j \in J'} \frac{R_j}{Q_v} \cdot x_{piv} \cdot (F_v \cdot (B_{pj} + B_{ijv} + B_{jp}) \cdot \tau + C_v)$$

$$TC3 = \sum_{t \in T} \sum_{\substack{n,o \in N \\ n \neq o}} \frac{S_{no}}{Q_t} \cdot Z_{not} \cdot (F_t \cdot B_{no} \cdot \tau + C_t) + \sum_{t \in T} \sum_{\substack{p,u \in P \\ p \neq u}} \frac{S_{pu}}{Q_t} \cdot Z_{put} \cdot (F_t \cdot B_{pu} \cdot \tau + C_t)$$

$$AC = \sum_{n \in N} Y_n + \left(\theta \cdot \sum_{i \in I} R_i \right) + \sum_{p \in P} Y_p + \left(\theta \cdot \sum_{j \in J} R_j \right) - \sum_{h \in N \cup P} \gamma_h \cdot Z_{np}$$

Therefore, the model formulation is as follows:

$$\min(F1 = TC1 + TC2 + TC3 + AC) \quad (1)$$

$$\min \left(F2 = \sum_{v \in V} \sum_{i \in I} \frac{R_i}{Q_v} \cdot x_{niv} + \sum_{v \in V} \sum_{j \in J} \frac{R_j}{Q_v} \cdot x_{piv} + \max \left\{ \sum_{v \in V} \sum_{i \in I} \frac{R_i}{Q_v}, \sum_{v \in V} \sum_{j \in J} \frac{R_j}{Q_v} \right\} \cdot x_{ijv} \right) \quad (2)$$

subject to

$$\left(\sum_{i \in I} R_i^* + \sum_{j \in J} R_j^* \right) \cdot x_{ijv} \leq Q_v \quad \forall v \in V \quad (3)$$

$$\left\{ \sum_{i \in I} R_i \cdot x_{niv}, \sum_{j \in J} R_j \cdot x_{piv} \right\} \leq Q_v \quad \forall v \in V \quad (4)$$

$$\left\{ \sum_{\substack{n,o \in N \\ n \neq o}} S_{no} \cdot Z_{not}, \sum_{\substack{p,u \in P \\ p \neq u}} S_{pu} \cdot Z_{put} \right\} \leq Q_t \quad \forall t \in T \quad (5)$$

$$\sum_{i \in I} x_{niv} = 1 \quad \forall n \in N, \forall v \in V \quad (6)$$

$$\sum_{i,j \in D} x_{ijv} - \sum_{j \in J} x_{jp} = 0 \quad \forall v \in V \quad (7)$$

$$\sum_{v \in V} \sum_{i \in I} R_i \cdot x_{niv} \leq Q_n \quad \forall n \in N \quad (8)$$

$$\sum_{v \in V} \sum_{i \in I} R_j \cdot x_{piv} \leq Q_p \quad \forall p \in P \quad (9)$$

$$\sum_{o \in N} \sum_{i \in I} x_{oiv} \leq 1 \quad \forall v \in V \quad (10)$$

$$\sum_{u \in P} \sum_{j \in J} x_{ujv} \leq 1 \quad \forall v \in V \quad (11)$$

$$\theta \geq 0 \quad (12)$$

$$Z_{not} = \{0, 1\} \quad \forall n, o \in N, t \in T \quad (13)$$

$$x_{niv} = \{0, 1\} \quad \forall n \in N, i \in I, v \in V \quad (14)$$

$$Z_{put} = \{0, 1\} \quad \forall p, u \in P, t \in T \quad (15)$$

where

$$x_{piv} = \{0, 1\} \quad \forall p \in P, j \in J, v \in V \quad (16)$$

$$Z_{np} = \{0, 1\} \quad \forall n \in N, p \in P \quad (17)$$

$$x_{ijv} = \{0, 1\} \quad \forall i \in I, j \in J, v \in V \quad (18)$$

The objectives of our research are represented in Eqs. (1)–(2) and respectively aim at finding minimum values of total operating cost and the size of deployed transportation resources. Constraints (3)–(5) define semitrailer truck and vehicle capacities. In constraint (3), we stipulate that the sum of remaining delivery and pickup quantity should not exceed the vehicle's maximum capacity. Constraints (4) and (5) assure the maximum capacity of semitrailer trucks and vehicles are higher or equal to the total demand of their assigned nodes. Constraint (6) implies that delivery vehicles always leave distribution centers. Constraint (7) is a flow conservation constraint which ensures that vehicles which simultaneously perform deliveries and pickups return to pickup centers. Constraints (8) and (9) guarantee that distribution and pickup centers have enough inventories to meet the total demand of their customers. Constraints (10) and (11) stipulate that delivery and pickup units should be served at most once. Constraints (12)–(18) state the non-negative and decision variables.

3. Research methodologies

3.1. Hybrid non-dominated sorting genetic algorithm-II

Vehicle routing optimization is known to be an NP-hard problem. As an extension of VRP, the CMCVRPSDP is also a complex problem, since the increase in the number of nodes in the logistics network leads to the increase of decision variables. In addition, the proposed problem propels transportation resource sharing, thereby requiring routes to be assigned accordingly. Owing to such complexity of the problem, using commercial solvers may constrain the interpretation of practical facts and yield less efficient results. In other words, applying heuristics proves to be better than commercial solvers to the extent that population-based algorithms offer a large range of feasible solutions. The Non-dominated Sorting Genetic Algorithm-II (NSGA-II) is a well-known evolutionary algorithm proposed by Deb et al. [14] and especially used to solve different types of multi-objective optimization problems [21,49]. As one of the best multi-objective optimization heuristics, NSGA-II suits with the proposed mathematical problem and can effectively be applied to evaluate the cost and number of vehicle objective functions. In addition, NSGA-II is adopted by many researchers for its fast elitist non-dominated sorting and its elitism at defining optimal solutions through the computation of crowding distances [1]. Therefore, developing a solution approach based on NSGA-II's framework is appropriate for the investigated CMCVRPSDP. Besides, customer clustering has also recently been adopted by scholars as a good strategy to decrease the computational complexity of optimization problems. For instance, Rabbani et al. [37] combined customer clustering with NSGA-II to address location-routing problems, and used proximity and vehicle capacity as clustering criteria. Their combination strategy is similar to that of this article, because we propose an optimization approach which is not only based on the framework of Deb's NSGA-II, but also combines k-means algorithm. The pseudo code of the proposed hybrid heuristic algorithm is presented below.

Pseudo code of the Hybrid Non-dominated Sorting Genetic Algorithm-II (HNSGA-II)

Step 1: Load pre-processed data; set the number k of clusters to form, the number r of runs, the population size $Popsiz$, the maximum number of generation $Gmax$ and genetic parameters

Step 2: if the coalition is only composed of DC or PC

Set k DC or PC as centers of corresponding clusters

Evaluate the distance between each delivery unit i or pickup unit j and each cluster center c , and assign i or j to the closest c
Repeat the previous step until every unit is assigned to the most appropriate DC or PC
else Set k as a vector where $k(1)$ designates the number of DC and $k(2)$ the number of PC
Select $k(1)$ DC and $k(2)$ PC as centers of corresponding clusters
Evaluate the distance between each delivery unit i and each DC, and the distance between pickup unit j and each center c , be it a DC or PC
Assign i to the closest DC and j to the closest c
Repeat the previous step until every unit is assigned to the most appropriate facility.
end; Output the allocation results of each facility in the coalition
for $c = 1:sum(k)$
for $run = 1:r$
Step 3: Generate the initial population
Step 4: Evaluate the objective functions and record the results
Step 5: Apply partial-mapped crossover (PMX) and swap mutation to generate the offspring population
Step 6: Combine the parent and offspring populations, and repeat step 4
Step 7: Conduct a non-dominated sorting and compute the crowding distance between chromosomes in the combined population
End
Step 8: Save the best solution of the run
End

The second step of the proposed hybrid heuristic consists in clustering customers into smaller and better manageable entities. The pseudo-code of HNSGA-II shows two different clustering procedures that are determined by the logistics network layout. In fact, the components of each coalition are decision variables to consider while undertaking clustering operations. On the one hand, a collaborative network with distribution centers should not consider the position of pickup centers during customer clustering, vice versa. In other words, only the number of DCs in a distribution network determines the number of clusters to form, whereas pickup networks consider the number of PCs. On the other hand, the formation of customer clusters in a collaborative network with both distribution and pickup centers is more complex. For the stability of the coalition, every facility has to be assigned customers; therefore, the number of clusters is equal to the sum of numbers of DCs and PCs.

Under both of scenarios, clusters centers are required to be logistics facilities location. Thus, instead of random initialization, this paper applies a controlled process to ensure the proper assignment of customers to each facility. Delivery and pickup nodes are primarily represented on a graph before the application of k-means. As with distance-based graph clustering approaches [7], k-means clustering algorithm is selected for its efficiency at addressing network environments where multiple nodes are interconnected. The application of different proximity functions is used to control the layout of the network. These functions do not only play an important role in the final configuration of the network but influence too the suitability of obtained results [31,45]. In the proposed clustering procedure, two proximity functions (Euclidean and cosine distance functions) are iteratively used, and results have been compared to select the most appropriate configuration. As a controlled procedure, our clustering approach may prove time-consuming when addressing networks with a large number of logistics facilities. But, in real world environment, coalitions are generally formed among limited numbers of members. Therefore, the proposed customer clustering procedure is entirely appropriate to the CMCVRPSDP optimization algorithm.

In summary, every customer is primarily assigned to a cluster according to its proximity with the center. The number of group is equal

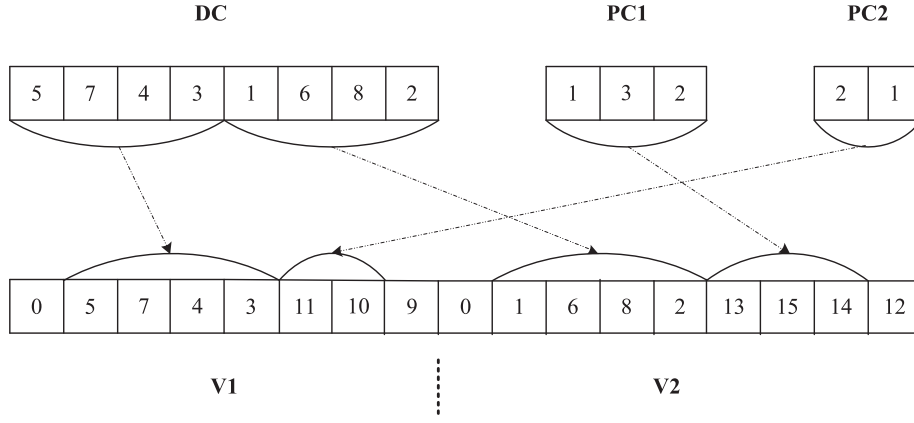


Fig. 2. Illustration of HNSGA-II chromosome encoding.

to the number of facilities in the corresponding coalition. Initial solutions in HNSGA-II are generated on the basis of clustering results. For example, consider a coalition composed of one DC with 8 delivery units and two PCs (PC1, PC2) with 10 pickup units each. After clustering, we suppose DC is assigned 3 units from PC1 and 2 from PC2. Considering vehicles capacity and units' demands, two vehicles can satisfy delivery and pickup requirements. Chromosomes in the initial population are encoded as in Fig. 2.

The distribution center is represented by number zero and its customers from one to eight. Pickup center PC2 is denoted by $I + 1$, with I being the total number of DC's customers, and its units obtained by adding $I + 1$ to the initial string. Pickup center PC1 is referred to as number $I + J_1 + 1$, with J_1 being the total number of PC2's customers, and its units obtained by adding $I + J_1 + 1$ to the corresponding initial string. The number of vehicles is defined by the total demand and vehicle's maximum capacity. As shown in Fig. 2, the first vehicle would visit four distribution nodes before joining nodes 11 and 10. As the collection of products is completed, V1 is constrained to return to the pickup center corresponding to the visited pickup units, hence PC1. It is important to note that vehicles are required to finish deliveries to every node on their routes before pickup operations.

Through the process described above, chromosomes are initialized and their fitness as solution to the CMCVRPSDP is later evaluated. Further, genetics operations are conducted to generate new candidate chromosomes. For that purpose, NSGA-II was originally designed with simulated binary crossover (SBX) and polynomial mutation (PoM) operators. But, in order to bind with the encoding scheme used in this paper and to ensure the satisfaction of the conditions allowing a single vehicle to perform both delivery and pickup, PMX and swap mutation operators have been applied instead of SBX and PoM, respectively. Before any genetic operation, generated chromosomes are preprocessed in order to assure that assumption (b) in Section 2.2 is bound, and that delivery vehicles should only collect from one pickup center's initial customers. The processing steps are as below:

Step 1: Find a vector denoted w of positions occupied by node 0. Every route starts from DC; therefore the first number in vector w is certainly "1". Aware of this fact, nodes at positions $[w(2:\text{end})-1, \text{end}]$ are recorded as boundaries of parent strings.

Step 2: Set sequences in-between as independent chromosomes and conduct PMX and swap mutation. The configuration of V2's route in Fig. 2 is represented by Parent 1 in Fig. 3.

The selection of PMX as crossover operator is meant to facilitate the transformation from the initial parent to an offspring where pickup nodes might be mixed with delivery nodes and yield a new chromosome as depicted in Fig. 3. A crossover region in two parents is selected and exchanged with one another, and the rest of the nodes are arbitrarily

arranged such that repetition is avoided. Further, swap mutation is conducted on a chromosome by randomly selecting two nodes and exchanging their position to form a different string. After crossover and mutation, re-insert nodes of vector w in their initial positions and recombine new strings to form a complete offspring chromosome for the evaluation of objective functions.

3.2. Profit allocation and coalition stability evaluation

In theory, profit allocation, cost sharing, coalition stability evaluation and coalition sequences selection are generally studied with the application of game theoretic methods [17,18]. A recent example is Joudia et al. [22] where a cooperative replenishment algorithm is proposed to search core-stable solutions and suggest horizontal collaboration as providing more cost saving than non-collaboration environments. This paper goes further to propose vehicle sharing as one of the coordinator negotiation terms in order to evaluate the effect of its implementation on logistics networks. After routes are properly designed, the total cost saving can be computed and profits can be allocated to each coalition's member. From savings computation to the final coalition's stability examination, the process is described in the following sub-sections.

3.2.1. Cost savings calculation

Through the network optimization, every logistics facility in a coalition can significantly save cost. The reduction of the total expenses (sum of warehousing, transportation and maintenance costs) is indeed a result of LSP negotiations with members of the group. Consequently, LSP would require a predetermined percentage of the collective profit as its coordination service payoff before distributing the remaining part to the participants. The corresponding percentage, also called the synergy requirement is defined by $\rho \in [0, 1]$. In practice, LSP can set a higher value for ρ to maximize profits, while members of the coalition would be more inclined to accept a lower ρ . For any value of the synergy requirement, cost savings are calculated using Eq. (19).

$$M(S) = \begin{cases} (1 - \rho)[\sum_{u \in S} C_0(u) - C(S)] & \text{if } \sum_{u \in S} C_0(u) - C(S) \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

$$R_{LSP} = \rho^* \left[\sum_{u \in S} C_0(u) - C(S) \right] \quad (20)$$

Where $C_0(u)$ expresses the initial cost of the player u without coalition. The corresponding cost saving of a coalition $M(S)$ is obtained as the gap between the initial cost $C_0(u)$ and the optimized cost $C(S)$. LSP receives as reward R_{LSP} and logistics facilities will share $M(S)$. However, if $\sum_{u \in S} C_0(u) - C(S) < 0$, members in coalition S will not accept the service provided by the LSP.

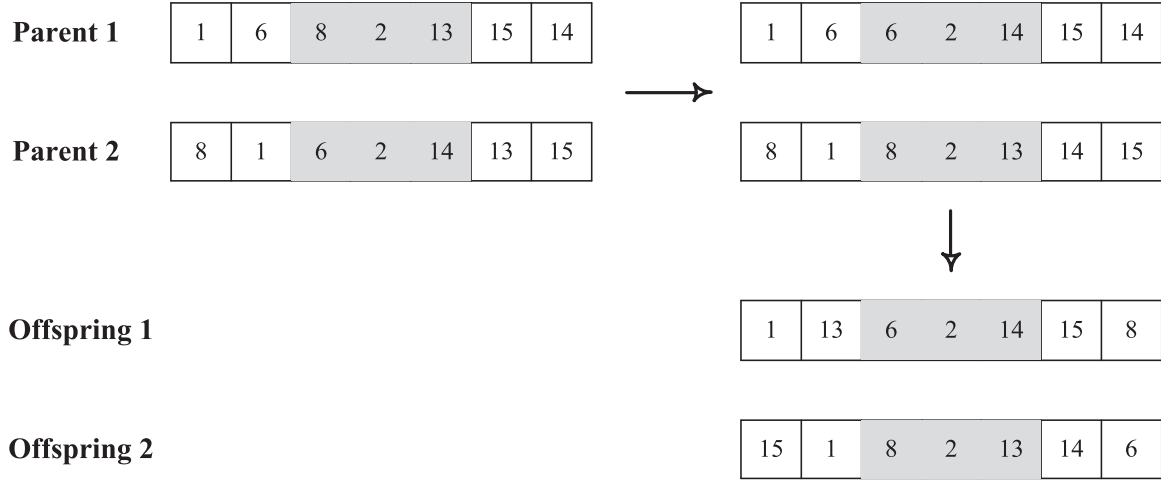


Fig. 3. Partial mapped crossover for CMCVRPSDP.

3.2.2. Cost gap allocation model

Cost Gap Allocation (CGA) is a cost-sharing method often used to solve profit allocation problems. CGA is based on the hypothesis that every player gains a certain benefit from any economic interaction with another entity. The model consists of two key concepts which are the marginal cost and the cost gap. Let $A = \{DC1, DC2, DC3, PC1, PC2\}$ represent the set all the players and S be a subset of A , which represents a type of coalition. The agreement of all the facilities to form a single coalition leads to the establishment of the grand coalition, whereas constituting multiple groups yields $(2^5 - 1)$ different subsets except null ones. Note that a single member can also be seen as a special coalition. In the cost allocation strategy, the marginal cost m_u^c represents the cost of player u while joining grand coalition A , and the cost gap in coalition S is denoted $g^c(S)$. Their respective formulas are given by Eqs. (21) and (22).

$$m_u^c = C(A) - C(A - \{u\}), \forall u \in A \quad (21)$$

$$g^c(S) = M(S) - \sum_{u \in S} m_u^c, \forall S \subset A, \quad (22)$$

$M(S)$ corresponds to the cost reduction after forming coalition S and $C(A)$ represents the total cost when the grand coalition is established. Subsequently, each player's profit can be calculated according to Eq. (23).

$$Z_u = m_u^c + \frac{\min_{u \in A} g^c(S)}{\min_{u \in A} g^c(S)} g^c(A), \forall u \in A \quad (23)$$

CGA model allocates profit or cost savings to each participant u . Among others, one of its advantages is related the simplicity of the model and the possibility to control the profit allocation process to ensure the stability of the coalition.

3.2.3. Strictly monotonic path definition

The best among all types of coalition is determined according to the monotonic path selection principle [10]. The increase in the cost reduction percentage as a new player joins the coalition is evaluated and scenarios are compared before the LSP decides which facilities to contact first. Let ω designate a sequence of players in the coalition; $\omega_{(l)}$ denotes the rank of player u in sequence ω . The cost reduction percentage $\eta(u, \omega, l)$ to player u on the l th step along sequence ω can be calculated using Eq. (24).

$$\eta(u, \omega, l) = Z_u \left(\bigcup_{\omega_{(q)} \leq l, q \in A} q, M \right) / C_0(u) \quad (24)$$

Sequences respecting the strictly monotonic paths (SMP) principle

are considered as optimal schemes. The selection of SMP-based sequences is a three-step process. (i) Select the values on the diagonal of SMP-based coalitions' cost reduction percentage matrix. (ii) Choose the participant with the maximal lowest reduction, or the second-in-order when the percentages are equal. Repeat until every participant is assessed or at least one is selected. (iii) Set the resulting sequence as a candidate to the best coalition sequence. Requirements are met if all the participants have been searched.

4. Empirical analyses

4.1 Network illustration and Vehicle routing optimization

The applicability of the proposed CMCVRPSDP is studied a logistics network in Chongqing city. Chongqing is located in the southwestern part of China and is acknowledged as an important logistics and transportation hub in the country. Three DCs (DC1, DC2, DC3), two PCs (PC1, PC2), and 128 representative customers, as illustrated in Fig. 4 and presented in Table 2 are assessed. In the existing logistics network, nodes in green are assigned to DC1, DC2 serves customers in yellow, red colored nodes are allocated to DC3, PC1's pickup units are represented by squares and PC2's pickup units illustrated with circles. Distribution and pickup centers were respectively engaged in separate delivery and pickup. But, considering the location of their respective clients, the initial (non-collaborative) network contains complicated links between logistics facility and customer, driving thus the entire system to low operational efficiency. To improve the network's reliability and attain sustainability, this article proposes collaboration between logistics facilities to optimize the initial network.

As mentioned in the model formulation, the objectives of our CMCVRPSDP optimization are the find routing solutions with minimum operating cost and number of vehicles. We have executed all computations on Matlab R2015a software. Parameters set during the initialization of HNSGA-II are defined after consultation of existing literature [46]. For each computation, we define: $C_t = 500$; $C_v = 100$; $\tau = 4.5$; $\lambda = 0.15$; $Q_t = 1800$; $Q_v = 200$; $F_t = 6.6$; $F_v = 3.2$. For fixed cost and discount offered to the each participant, distribution and pickup centers are indexed such that the set $\{DC1; DC2; DC3; PC1; PC2\}$ corresponds to $\{1; 2; 3; 4; 5\}$. Therefore, $Y_1 = 1200$; $Y_2 = 1000$; $Y_3 = 900$; $Y_4 = 1700$; $Y_5 = 2000$. Incentives offered by LSP are: $\gamma_1 = 1000$; $\gamma_2 = 850$; $\gamma_3 = 765$; $\gamma_4 = 1300$; $\gamma_5 = 1600$. Referring to the algorithm parameters, many tests are conducted using HNSGA-II and results are compared to calibrate the population size Pop_{size} , the number of generation max_{gen} , the crossover probability P_c and the mutation probability P_m . The most appropriate calibration is as follows: $Pop_{size} = 150$; $max_{gen} = 1000$; $P_c = 0.85$; $P_m = 0.1$.

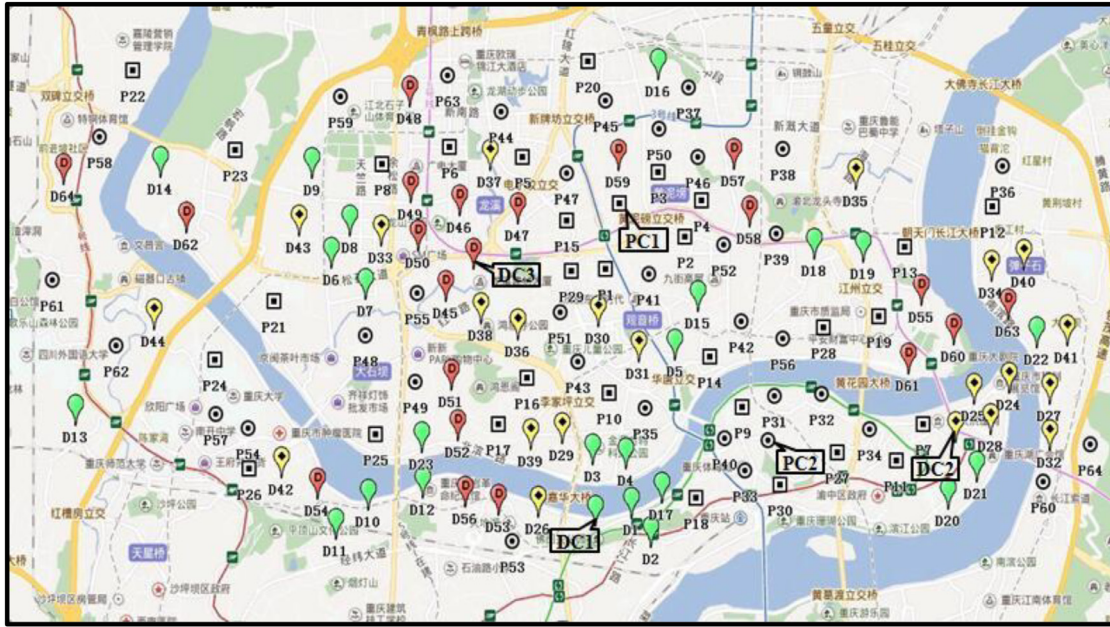


Fig. 4. Geographical distribution of customers, distribution and pickup centers.

Following 10 runs for each scenario, optimization results are summarized in Table 3 and illustrated in Fig. 5. In all circumstances, we observe significant reduction in the network operating cost and the number of vehicle dispatched for delivery and pickup activities. Thus, in order to minimize expenses, logistics facilities should cooperate and share their corresponding transportation resources. In addition, Fig. 5 displays that largest gaps between cost solutions and numbers of vehicle are yielded when the coalition regroupes every facility of the network. Although collaborating with any other facility is cost effective, distribution and pickup centers would increase their profit by joining the grand coalition. Nevertheless, this inference still needs complementary analyses to confirm its relevance. Therefore, Section 4.3 provides an insightful evaluation of the vehicle routing cost solutions to provide more valuable conclusions.

Moreover, vehicle sharing constitutes one of the major advantages of CMCVRP and a factor supporting the quality of optimized solutions. Through the use of delivery vehicles to perform pickups, the grand coalition has reduced the total number of vehicles from 13 to 10. This three-vehicle gap also accounts for the minimization of the total expense, thereby offering more benefit for a stable profit distribution.

4.2. Evaluation of algorithms performance

In order to verify the relevance of the proposed HNSGA-II as an innovative algorithm, we have conducted series of comparison with well-known heuristics by evaluating their capability at solving the case study as well as multiple depots vehicle routing problems in general. Some researchers demonstrated the outperformance of NSGA-II on many other heuristics [1,3]. But, since the proposed hybrid heuristic

includes some modifications on the original NSGA-II, it is worth conducting a comparison to evaluate the effects of our improvements. Besides, we also apply the Multi-objective particle swarm optimization (MOPSO) algorithm to find effective routing solutions. Widely known for its capacity to perform solutions local search, MOPSO is selected to show the ability of clustering at improving solution quality [35].

On the one hand, algorithms performance is evaluated on the investigated case study. Each algorithm is used to operate 10 separate runs and comparative results are summarized in Table 4. HNSGA-II yields the minimum cost for every run except the third one where NSGA-II comes first. Though this solution is dominated, the existence of the lowest cost for the remaining runs and the smallest average cost solution are sufficient reasons to enhance its good performance. In addition, a significant number of runs have solutions closer or equal to \$2900 which is the minimum cost solution, and thereby supports the convergence of HNSGA-II's optimization results. Besides, the number of vehicles remains constant for every algorithm throughout all the 10 runs. Therefore, 10 vehicles are the optimal number of vehicles in the collaborative network grouping DC1, DC2, DC3, PC1 and PC2. In summary, compared to NSGA-II and MOPSO, HNSGA-II is the most appropriate heuristic algorithm to assess the current case study because of the ability at providing the lowest cost solution and smallest number of vehicles.

On the other hand, modified Solomon datasets are employed to evaluate the performance of HNSGA-II on benchmarks. For comparative purpose, the total operating cost, number of vehicles, and the algorithm running time are considered as performance measures. Table 5 summarizes optimization results corresponding to each instance.

Optimization results show that besides instances 14, 22, 23, 25, and

Table 2
Initial customer allocation plan.

Facilities	Allocated customers
DC1	D1; D2; D3; D4; D5; D6; D7; D8; D9; D10; D11; D12; D13; D14; D15; D16; D17; D18; D19; D20; D21; D22; D23.
DC2	D24; D25; D26; D27; D28; D29; D30; D31; D32; D33; D34; D35; D36; D37; D38; D39; D40; D41; D42; D43; D44.
DC3	D45; D46; D47; D48; D49; D50; D51; D52; D53; D54; D55; D56; D57; D58; D59; D60; D61; D62; D63; D64.
PC1	P1; P2; P3; P4; P5; P6; P7; P8; P9; P10; P11; P12; P13; P14; P15; P16; P17; P18; P19; P20; P21; P22; P23; P24; P25; P26; P27; P28; P29; P30.
PC2	P31; P32; P33; P34; P35; P36; P37; P38; P39; P40; P41; P42; P43; P44; P45; P46; P47; P48; P49; P50; P51; P52; P53; P54; P55; P56; P57; P58; P59; P60; P61; P62; P63; P64.

Table 3
Optimization results comparison.

Coalitions	Initial network		Optimized network	
	Cost	Vehicle	Cost	Vehicle
[DC1]	1690	3	1131	3
[DC2]	1348	2	890	2
[DC3]	1359	3	922	3
[PC1]	2087	2	1324	2
[PC2]	2542	3	1628	3
[DC1 PC1]	3777	5	1549	4
[DC1 PC2]	4232	6	1797	5
[DC2 PC1]	3435	4	1300	4
[DC2 PC2]	3890	5	1686	4
[DC3 PC1]	3446	5	1356	5
[DC3 PC2]	3901	6	1472	5
[DC1 DC2]	3038	5	1693	5
[DC1 DC3]	3049	6	1776	6
[DC2 DC3]	2707	5	1612	5
[PC1 PC2]	4629	5	2062	5
[DC1 DC2 PC1]	5125	7	2279	6
[DC1 DC2 PC2]	5580	8	2573	8
[DC1 DC3 PC1]	5136	8	2364	7
[DC1 DC3 PC2]	5591	9	2448	7
[DC2 DC3 PC1]	4794	7	2098	6
[DC2 DC3 PC2]	5249	8	2284	6
[DC1 PC1 PC2]	6319	8	2641	7
[DC2 PC1 PC2]	5977	7	2472	6
[DC3 PC1 PC2]	5988	8	2560	7
[DC1 DC2 DC3]	4397	8	2183	7
[DC1 DC2 PC1 PC2]	7667	10	3607	8
[DC1 DC3 PC1 PC2]	7678	11	3741	10
[DC2 DC3 PC1 PC2]	7336	10	3574	9
[DC1 DC2 DC3 PC1]	6484	10	2770	8
[DC1 DC2 DC3 PC2]	6939	11	2855	8
[DC1 DC2 DC3 PC1 PC2]	9026	13	2900	10

27, HNSGA-II performs the best in the remaining 26 instances. In average, our approach yields minimum cost and vehicle solutions using the smallest amount of running time in every computation. Though the 23rd instance displays HNSGA-II's cost as the maximum, the difference with respect to the best answer is of 3.8%. In the same instance, HNSGA-II obtains the lowest number of vehicles and has a significant advantage in term of computation time. As for the remaining instances where the proposed algorithm yields the second-best result, it is important to highlight that instance 22 has the best solution in term of number of vehicles. In summary: (i) for none of the instances has HNSGA-II obtained the worst result in all the measures; (ii) HNSGA-II is faster than NSGA-II and MOGA on every instance; (iii) On small size instances, HNSGA-II possesses an absolute superiority since number of

Table 4
Algorithms comparison on the investigated case study.

Runs	HNSGA-II		NSGA-II		MOPSO	
	Cost	Vehicle	Cost	Vehicle	Cost	Vehicle
1	3241	10	3253	10	3301	10
2	3240	10	2995	10	3295	10
3	2900	10	3295	10	3348	10
4	3002	10	3256	10	3012	10
5	2909	10	2953	10	2994	10
6	3240	10	3256	10	3304	10
7	2901	10	2922	10	3206	10
8	2930	10	3297	10	2970	10
9	2900	10	2946	10	3304	10
10	3241	10	3256	10	3299	10
Average	3084	10	3143	10	3233	10

vehicles and running time results are the best for instance 1–13 with 10 to 50 customers; (iv) though some medium size benchmarks show a lower performance for HNSGA-II, the majority of the cases and the average values for instances 14–30 still supports an outperformance. Therefore, we can conclude that HNSGA-II displays superiority over NSGA-II and MOGA and is the best among the three algorithms at optimizing small and medium sized networks.

On the basis of the results in Tables 4 and 5, the following observations can also be drawn. (i) Although particle swarm optimization algorithms are better than genetic algorithms at searching local optima, NSGA-II ensures the quality of its solution by non-dominated sorting feasible solutions. Besides, NSGA-II is a population-based heuristics, with the wide range of initially generated solutions, its fast elitist non-dominated sorting and crowding distance assignment operations increase the probability of finding better solutions. (ii) Customer clustering is a process which improves algorithm performance. In contrast with the typical NSGA-II, the integration of k-means clustering algorithm in our proposed hybrid heuristic has a positive impact on solution quality. By reducing the number of long trips, clustering customers plays an important role by minimizing the average operating cost.

4.3. Profit allocation

In practice, logistics facilities adhesion to any coalition is subject to the assurance of gaining profit from the entire network's operation. The majority of logistics company strategies are economically driven and the more benefit is offered by the coordinator. Based on practical observation, assume LSP accepts 12% of the total cost saving, and set the synergy requirement as $\rho = 0.12$.

The remaining profit is allocated fairly among DC1, DC2, DC3,

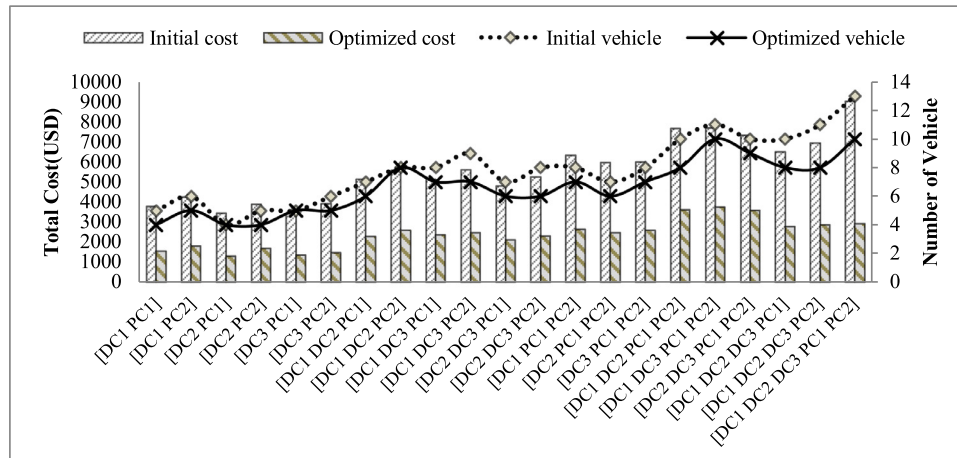


Fig. 5. Comparison of cost and number of vehicle solutions.

Table 5
Comparison of algorithms performances on benchmark instances.

Inst	N. of Customer	N. of Depot	HNSGA-II			NSGA-II			MOPSO		
			Cost (\$)	N. of Vehicle	Time (s)	Cost (\$)	N. of Vehicle	Time (s)	Cost (\$)	N. of Vehicle	Time (s)
1	10	2	1040	2	30	1102	2	37	1092	2	47
2	10	3	1536	3	35	1587	3	43	1583	3	62
3	20	2	2156	4	38	2308	4	52	2258	4	57
4	20	3	2447	4	43	2473	4	56	2504	4	59
5	20	4	2488	4	49	2518	4	63	2710	5	66
6	30	2	3759	6	46	3930	6	72	3963	6	75
7	30	3	3996	6	51	4037	6	77	4391	7	72
8	30	4	4398	6	56	4671	6	82	4846	7	81
9	30	5	4220	6	62	4408	6	89	4492	7	87
10	50	2	7326	9	67	7669	9	131	9897	10	109
11	50	3	8791	10	77	8818	10	132	9804	10	108
12	50	4	7447	9	73	7912	9	135	9055	11	105
13	50	5	7813	9	79	8349	9	185	9883	11	111
14**	70	4	16,408	14	94	16,060	13	211	16,893	15	146
15	70	6	12,000	14	101	12,431	14	220	14,577	16	141
16	70	8	10,222	14	111	11,662	14	232	12,534	15	145
17	70	10	11,034	15	123	11,336	15	246	13,403	18	159
18	70	12	13,562	15	134	13,717	15	255	15,015	18	174
19	90	4	20,380	17	119	21,255	17	425	22,203	18	369
20	90	6	20,816	18	124	21,079	18	411	21,262	19	350
21	90	8	24,897	17	138	25,211	18	423	26,213	20	411
22**	90	10	24,220	18	154	24,648	19	440	24,210	21	421
23*	90	12	30,553	19	174	29,541	20	462	29,426	22	441
24	100	4	21,120	19	141	22,704	20	492	25,333	21	407
25**	100	6	30,427	20	143	28,913	19	490	31,365	21	370
26	100	8	28,299	19	132	29,069	20	497	29,867	22	356
27**	100	10	33,444	20	141	32,753	19	394	36,740	24	363
28	100	12	30,811	20	150	30,938	21	392	35,369	24	371
29	100	14	33,731	23	161	36,893	23	405	38,989	25	400
30	100	16	33,569	22	171	34,366	22	413	36,258	24	413
Average			14,564	13	101	15,412	13	252	16,538	14	216

Note: ** correspond to instances where HNSGA-II yields second best results. * corresponds to instances where HNSGA-II obtains the maximum cost. Time in Table 5 refers to the unit running time in seconds.

Table 6
Collaborative network profit allocation.

S	$\sum_{u \in S} C_o(u)$	C(S)	M(S)	Facilities allocated profit
[DC1]	1690	1131	492	(492; ■; ■; ■)
[DC2]	1348	890	403	(■; 403; ■; ■)
[DC3]	1359	922	385	(■; ■; 385; ■)
[PC1]	2087	1324	671	(■; ■; ■; 671; ■)
[PC2]	2542	1628	804	(■; ■; ■; 804)
[DC1 PC1]	3777	1549	1961	(1122; ■; ■; 839; ■)
[DC1 PC2]	4232	1797	2143	(1050; ■; ■; ■; 1093)
[DC2 PC1]	3435	1300	1879	(■; 1004; ■; 875; ■)
[DC2 PC2]	3890	1686	1940	(■; 872; ■; ■; 1068)
[DC3 PC1]	3446	1356	1839	(■; ■; 930; 909; ■)
[DC3 PC2]	3901	1472	2138	(■; ■; 917; ■; 1221)
[DC1 DC2]	3038	1693	1184	(669; 515; ■; ■; ■)
[DC1 DC3]	3049	1776	1120	(691; ■; 429; ■; ■)
[DC2 DC3]	2707	1612	964	(■; 536; 428; ■; ■)
[PC1 PC2]	4629	2062	2259	(■; ■; ■; 967; 1292)
[DC1 DC2 PC1]	5125	2279	2504	(980; 826; ■; 698; ■)
[DC1 DC2 PC2]	5580	2573	2646	(919; 765; ■; ■; 962)
[DC1 DC3 PC1]	5136	2364	2439	(995; ■; 732; 712; ■)
[DC1 DC3 PC2]	5591	2448	2766	(995; ■; 733; ■; 1038)
[DC2 DC3 PC1]	4794	2098	2372	(■; 870; 762; 740; ■)
[DC2 DC3 PC2]	5249	2284	2609	(■; 840; 732; ■; 1037)
[DC1 PC1 PC2]	6319	2641	3237	(1159; ■; ■; 876; 1202)
[DC2 PC1 PC2]	5977	2472	3084	(■; 1005; ■; 876; 1203)
[DC3 PC1 PC2]	5988	2560	3017	(■; ■; 911; 890; 1216)
[DC1 DC2 DC3]	4397	2183	1948	(788; 634; 526; ■; ■)
[DC1 DC2 PC1 PC2]	7667	3607	3573	(992; 838; ■; 709; 1034)
[DC1 DC3 PC1 PC2]	7678	3741	3465	(992; ■; 730; 709; 1034)
[DC2 DC3 PC1 PC2]	7336	3574	3311	(■; 838; 730; 709; 1034)
[DC1 DC2 DC3 PC1]	6484	2770	3268	(992; 838; 730; 708; ■)
[DC1 DC2 DC3 PC2]	6939	2855	3594	(992; 838; 730; ■; 1034)
[DC1 DC2 DC3 PC1PC2]	9026	2900	5391	(1210; 1055; 947; 927; 1252)

PC1 and PC2 employing CGA model. All non-empty coalitions' profit distributions out of CGA application are shown in Table 6 (Unit: USD). Before collaboration, DC1, DC2, DC3, PC1, PC2 respectively gain as profit 492, 403, 385, 671 and 804, and in the grand coalition, their profits become 1210, 1055, 947, 927 and 1252, respectively. Results in Table 6 also indicate that every distribution center obtains the highest profit by joining the grand coalition, but pickup centers yield most when they collaborate between pickup centers. In other words, LSP would have to negotiate with PC1 and PC2, and convince them on the relevance of resource sharing and the advantage of the grand coalition.

To verify the accuracy of our profit distribution, the Minimum Costs-Remaining Savings (MCRS) method, Shapley method and Game Quadratic Programming (GQP) are applied to find distribution schemes corresponding to the grand coalition and compare them with CGA results. As shown in Table 7, their respective solutions are different. While PC1 and PC2 receive more according to Shapley method, CGA's values for the three DCs are the highest. Determining the best method among the proposed ones is therefore critical to find the optimal profit of each facility. The core concept is a generally used approach to verify profit allocation strategies [26,39]. Since our network contains five logistics facilities, the core is designed as a closed convex polyhedron.

Table 7
Comparison of profit distribution methods.

	Shapley	MCRS	GQP	CGA
DC1	1013	1091	1183	1210
DC2	884	977	1029	1055
DC3	860	925	921	947
PC1	1208	1097	966	927
PC2	1426	1301	1292	1252

Table 8
Core distance calculation and comparison.

Methods	Schemes	Center	Distance
MCRS	(1091;977;925;1097;1301)	(1142;1061;1000;	194
CGA	(1210;1055;947;927;1252)	1003;1185)	133
Shapley	(1013;884;860;1208;1426)		409
GQP	(1183;1029;921;966;1292)		148

The core center is obtained by compressing the polyhedron and narrowing the core into a single point. A variable α is used to control its range. Considering the marginal cost, the center is determined following Eq. (25).

$$\frac{C(A) - C(A - \{\beta\})}{C(A)} * \alpha + \sum_{a \in A} y_a = C(A - \{\beta\}) \quad (25)$$

The proximity of the profit allocation solutions to the core center determines the stability of the coalition [28]. As shown in Table 8, the center of the core in the investigated case is (1142, 1061, 1000, 1003, 1185) with $\alpha = -2405$.

Comparisons of the distance between each profit allocation method and the core center proves that the Shapley value is the farthest from the center, and DC1, DC2 and DC3 receive the lowest profits. Therefore, their possibility of refusing the offer is higher, and the stability of the coalition would be threatened. Consequently, Shapley's allocation plan has to be rejected. Similarly, solutions of the GQP and MCRS should be ignored because their distances are not the best. Compared with other models, CGA obtains a solution closer to the center and establishes stronger coalition stability. Therefore, CGA is the most appropriate allocation method in this study to fairly distribute profit among members of the grand coalition.

4.5. Sequential coalition selection

Synergy between multiple players primarily seeks to maximize profits. However, differences in adhesion sequences lead to differences in allocated profits and also influence the coalition stability. Consequently, the analysis of the impacts of sequential combinations on the profit allocation strategy allows decision makers to observe all the possible scenarios. In this regard, CGA results and the SMP principle are

used to evaluate each coalition formation sequence. The cost reduction percentages in all possible sequences are shown in Fig. 6. The trend of costs reduction percentages for coalitions respecting the SMP principle is presented in Tables 9–11.

Based on the diagonal values of each table, we conclude that the optimal sequence of Tables 9–11 are respectively {DC1, DC2, DC3, PC2, PC1}, {DC2, DC1, DC3, PC2, PC1} and {DC3, DC1, DC2, PC2, PC1}. Comparisons of these three sequences indicate that the optimal coalition of the collaborative network is $\omega = \{DC2, DC1, DC3, PC2, PC1\}$. In other words, the best collaboration strategy is described as follows: DC2 first joins the coalition and gets 34.0% cost reduction, DC1 follows to yield 38.2% and lead DC2 to 39.6% reduction; when DC3 joins, the reduction sequence becomes 47.0%, 46.6% and 38.7% respectively for DC2, DC1 and DC3; PC2 adheres and reduces members costs by 62.1%, 58.7%, 53.7% and 40.7%; finally as PC1 moves in, DC2, DC1, DC3, PC2, PC1 can reduce their costs by 78.3%, 71.6%, 69.7%, 49.3% and 44.4%, respectively. In summary, whenever a new facility joins the coalition, existing members do not only benefit from the efforts of the new member, but their efforts provide the new member with consistent cost reduction.

5. Discussion

This article mainly focuses on the analysis of a grand coalition. But in reality, logistics facilities can form more than one coalition. Assuming that a group should at least include two members, the maximum number of coalition to simultaneously exist in the network is two. Fig. 7(a) graphically illustrates the change in percentage for the optimal grand coalition sequence. Table 12 and Fig. 7(b) discuss scenarios where the entire network operates with two sub-coalitions.

According to the diagonal rule in the snowball theory, optimal sequential coalitions are $\omega_1 = \{PC1, DC2\}$, $\omega_2 = \{DC1, DC3, PC2\}$. PC1 and DC2 form a coalition, whereas DC1, DC3 and PC2 form another one. In ω_1 , PC1 first enters the coalition and obtains 32.2% cost reduction; DC2 joins to yield final reduction percentages of 41.9% and 74.5% respectively for PC1 and DC2. In ω_2 , DC1 is first invited and receives 29.1% cost reduction; after PC2 enters the second coalition, DC1, DC3 and PC2 respectively obtain 58.9%, 53.9% and 40.8% reduction.

The formation of the grand coalition offers distribution and pickup

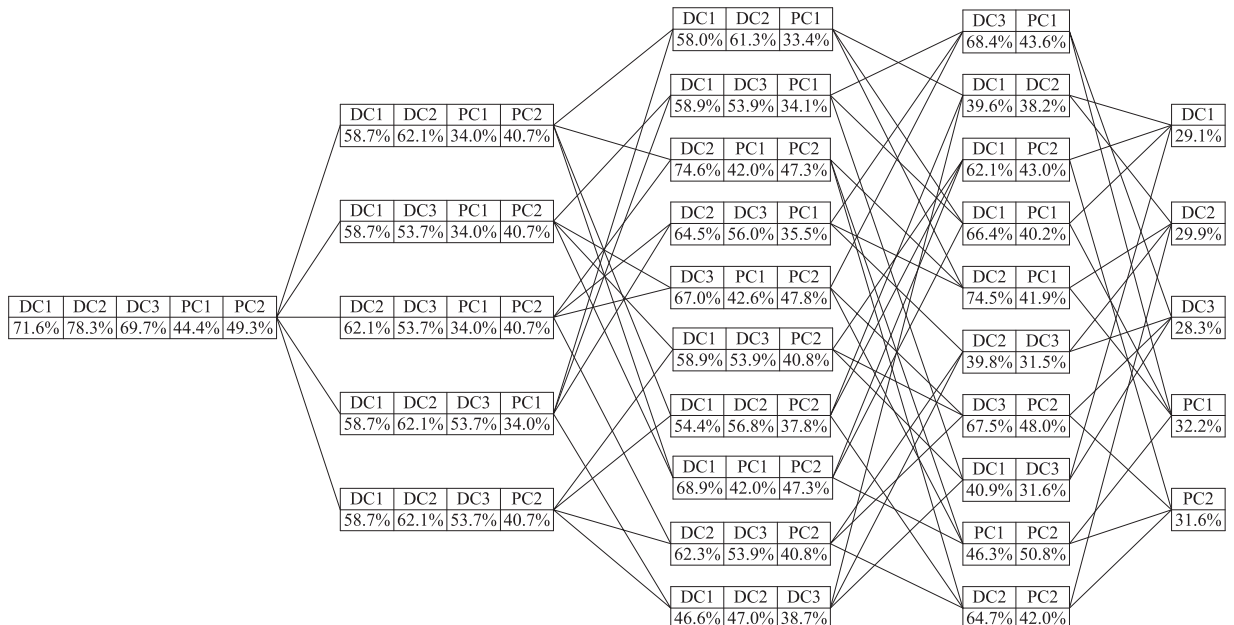


Fig. 6. Cost reduction percentages for all coalitions.

Table 9
Sequential coalitions starting from DC1.

$\omega_1 = \{DC1, DC2, DC3, PC2, PC1\}$						$\omega_2 = \{DC1, DC2, DC3, PC1, PC2\}$					
Player u	DC1	DC2	DC3	PC2	PC1	Player u	DC1	DC2	DC3	PC1	PC2
$\eta(u, \omega, 1)$	29.1%					$\eta(u, \omega, 1)$	29.1%				
$\eta(u, \omega, 2)$	39.6%	38.2%				$\eta(u, \omega, 2)$	39.6%	38.2%			
$\eta(u, \omega, 3)$	46.6%	47.0%	38.7%			$\eta(u, \omega, 3)$	46.6%	47.0%	38.7%		
$\eta(u, \omega, 4)$	58.7%	62.1%	53.7%	40.7%		$\eta(u, \omega, 4)$	58.7%	62.1%	53.7%	34.0%	
$\eta(u, \omega, 5)$	71.6%	78.3%	69.7%	49.3%	44.4%	$\eta(u, \omega, 5)$	71.6%	78.3%	69.7%	44.4%	49.3%
$\omega_3 = \{DC1, DC2, PC2, PC1, DC3\}$						$\omega_4 = \{DC1, DC2, PC2, DC3, PC1\}$					
Player u	DC1	DC2	PC2	PC1	DC3	Player u	DC1	DC2	PC2	DC3	PC1
$\eta(u, \omega, 1)$	29.1%					$\eta(u, \omega, 1)$	29.1%				
$\eta(u, \omega, 2)$	39.6%	38.2%				$\eta(u, \omega, 2)$	39.6%	38.2%			
$\eta(u, \omega, 3)$	54.4%	56.8%	37.8%			$\eta(u, \omega, 3)$	54.4%	56.8%	37.8%		
$\eta(u, \omega, 4)$	58.7%	62.1%	40.7%	34.0%		$\eta(u, \omega, 4)$	58.7%	62.1%	40.7%	53.7%	
$\eta(u, \omega, 5)$	71.6%	78.3%	49.3%	44.4%	69.7%	$\eta(u, \omega, 5)$	71.6%	78.3%	49.3%	69.7%	44.4%

Table 10
Sequential coalitions starting from DC2.

$\omega_1 = \{DC2, DC1, DC3, PC2, PC1\}$						$\omega_2 = \{DC2, DC1, DC3, PC1, PC2\}$					
Player u	DC2	DC1	DC3	PC2	PC1	Player u	DC2	DC1	DC3	PC1	PC2
$\eta(u, \omega, 1)$	29.9%					$\eta(u, \omega, 1)$	29.9%				
$\eta(u, \omega, 2)$	38.2%	39.6%				$\eta(u, \omega, 2)$	38.2%	39.6%			
$\eta(u, \omega, 3)$	47.0%	46.6%	38.7%			$\eta(u, \omega, 3)$	47.0%	46.6%	38.7%		
$\eta(u, \omega, 4)$	62.1%	58.7%	53.7%	40.7%		$\eta(u, \omega, 4)$	62.1%	58.7%	53.7%	34.0%	
$\eta(u, \omega, 5)$	78.3%	71.6%	69.7%	49.3%	44.4%	$\eta(u, \omega, 5)$	78.3%	71.6%	69.7%	44.4%	49.3%
$\omega_3 = \{DC2, DC1, PC2, PC1, DC3\}$						$\omega_4 = \{DC2, DC1, PC2, DC3, PC1\}$					
Player u	DC2	DC1	PC2	PC1	DC3	Player u	DC2	DC1	PC2	DC3	PC1
$\eta(u, \omega, 1)$	29.9%					$\eta(u, \omega, 1)$	29.9%				
$\eta(u, \omega, 2)$	38.2%	39.6%				$\eta(u, \omega, 2)$	38.2%	39.6%			
$\eta(u, \omega, 3)$	56.8%	54.4%	37.8%			$\eta(u, \omega, 3)$	56.8%	54.4%	37.8%		
$\eta(u, \omega, 4)$	62.1%	58.7%	40.7%	34.0%		$\eta(u, \omega, 4)$	62.1%	58.7%	40.7%	53.7%	
$\eta(u, \omega, 5)$	78.3%	71.6%	49.3%	44.4%	69.7%	$\eta(u, \omega, 5)$	78.3%	71.6%	49.3%	69.7%	44.4%

Table 11
Sequential coalitions starting from DC3.

$\omega_1 = \{DC3, DC1, DC2, PC2, PC1\}$						$\omega_2 = \{DC3, DC2, DC1, PC1, PC2\}$					
Player u	DC3	DC1	DC2	PC2	PC1	Player u	DC3	DC2	DC1	PC1	PC2
$\eta(u, \omega, 1)$	28.3%					$\eta(u, \omega, 1)$	28.3%				
$\eta(u, \omega, 2)$	31.6%	40.9%				$\eta(u, \omega, 2)$	31.5%	39.8%			
$\eta(u, \omega, 3)$	38.7%	46.6%	47.0%			$\eta(u, \omega, 3)$	38.7%	47.0%	46.6%		
$\eta(u, \omega, 4)$	53.7%	58.7%	62.1%	40.7%		$\eta(u, \omega, 4)$	53.7%	62.1%	58.7%	34.0%	
$\eta(u, \omega, 5)$	69.7%	71.6%	78.3%	49.3%	44.4%	$\eta(u, \omega, 5)$	69.7%	78.3%	71.6%	44.4%	49.3%
$\omega_3 = \{DC3, DC1, DC2, PC1, PC2\}$						$\omega_4 = \{DC3, DC2, DC1, PC2, PC1\}$					
Player u	DC3	DC1	DC2	PC1	PC2	Player u	DC3	DC2	DC1	PC2	PC1
$\eta(u, \omega, 1)$	28.3%					$\eta(u, \omega, 1)$	28.3%				
$\eta(u, \omega, 2)$	31.6%	40.9%				$\eta(u, \omega, 2)$	31.5%	39.8%			
$\eta(u, \omega, 3)$	38.7%	46.6%	47.0%			$\eta(u, \omega, 3)$	38.7%	47.0%	46.6%		
$\eta(u, \omega, 4)$	53.7%	58.7%	62.1%	34.0%		$\eta(u, \omega, 4)$	53.7%	62.1%	58.7%	40.7%	
$\eta(u, \omega, 5)$	69.7%	71.6%	78.3%	44.4%	49.3%	$\eta(u, \omega, 5)$	69.7%	78.3%	71.6%	49.3%	44.4%

centers significant reduction on their operating cost and number of vehicles. Depending on their order of adhesion, every participant as well as the coordinator can expect a return equivalent to its contribution in the total profit of the coalition. Table 13 evaluates the best sequential coalitions between single and two coalition networks. Findings show that DC1, DC2, DC3, PC1 and PC2 respectively yield from the establishment of a grand coalition 78.3%, 71.6%, 69.7%, 49.3% and 44.4% cost reduction percentage. Under two sub-coalitions scenario, percentages are respectively 41.9%, 74.5%, 58.9%, 53.9% and 40.8%. In conclusion, a grand coalition is the best collective option for the five logistics facilities.

As shown in Fig. 8, in the grand coalition. We observe that cost reduction percentages are not proportional to expected profits. Examining PC1's results might prove the contrary, but the analysis of

every facility in the network supports our inference. For example, PC1 has the lowest cost reduction percentage (44.4%) and the smallest profit (\$927), whereas PC2's cost reduction displays the second lowest percentage (49.3%), but the highest profit (\$1252). PC2's profit can be due to the fact that it counts the highest number of customer and receives the largest discount, whereas the observations on PC1 can be supported by the traffic density on roads travelled by its vehicles. But in a long run, PC1 might take the number of its initial customers as a claim for profit allocation reconsideration, and threaten the stability of the grand coalition. Therefore, coordinators should evaluate factors like traffic conditions and roads quality in order to avoid potential internal matters. The promotion of information sharing among members of the same coalition can also reduce risks of instability and improve collaboration's reliability [30].

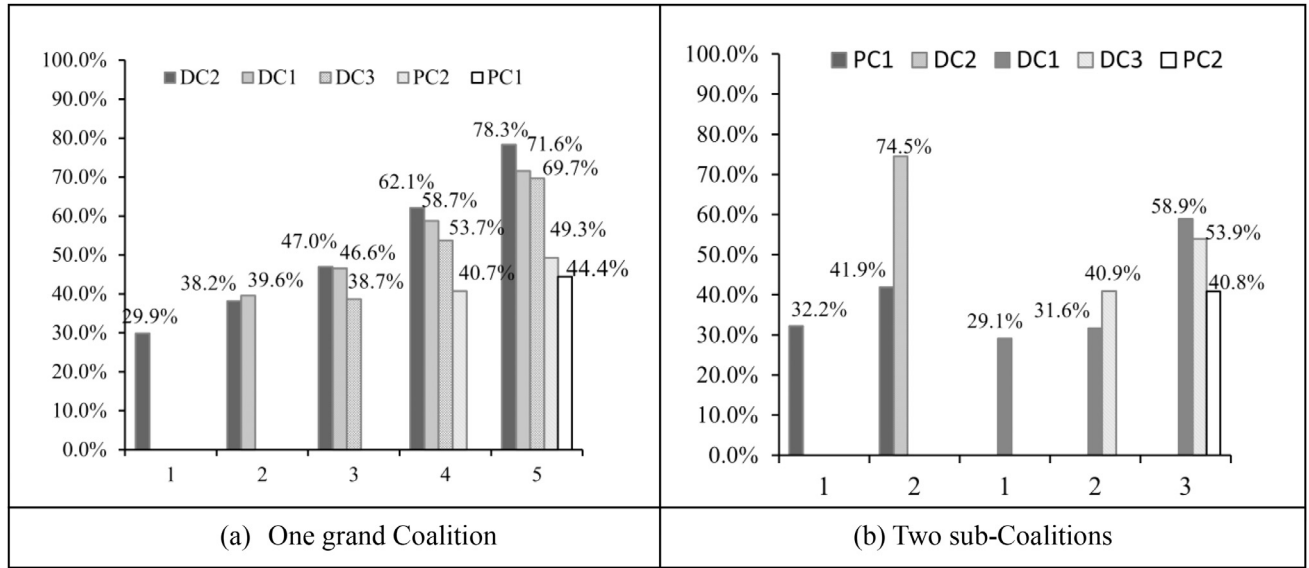


Fig. 7. Cost reduction percentage after collaboration.

Table 12

Optimal sequences for the formation of two sub-coalitions.

$\omega_1 = \{PC1, DC2\}$		$\omega_2 = \{DC1, DC3, PC2\}$				$\omega_1 = \{DC2, DC3\}$		$\omega_2 = \{DC1, PC2, PC1\}$			
Player u	PC1	DC2	DC1	DC3	PC2	Player u	DC2	DC3	DC1	PC2	PC1
$\eta(u, \omega, 1)$	36.6%		33.1%			$\eta(u, \omega, 1)$	34.0%		33.1%		
$\eta(u, \omega, 2)$	47.6%	84.6%	46.5%	35.8%		$\eta(u, \omega, 2)$	45.2%	35.8%	70.7%	48.8%	
$\eta(u, \omega, 3)$			66.9%	61.3%	46.4%	$\eta(u, \omega, 3)$			77.9%	53.7%	47.7%
$\omega_1 = \{PC2, DC2\}$		$\omega_2 = \{DC1, DC3, PC1\}$				$\omega_1 = \{PC1, PC2\}$		$\omega_2 = \{DC2, DC1, DC3\}$			
Player u	PC2	DC2	DC1	DC3	PC1	Player u	PC1	PC2	DC2	DC1	DC3
$\eta(u, \omega, 1)$	36.0%		33.1%			$\eta(u, \omega, 1)$	36.6%		34.0%		
$\eta(u, \omega, 2)$	47.7%	73.5%	46.5%	35.8%		$\eta(u, \omega, 2)$	52.7%	57.7%	43.4%	45.0%	
$\eta(u, \omega, 3)$			66.9%	61.3%	38.8%	$\eta(u, \omega, 3)$			53.5%	53.0%	43.9%

Table 13

Best sequences for one and two coalitions.

$\omega_1 = \{DC2, DC1, DC3, PC2, PC1\}$					
Player u	DC2	DC1	DC3	PC2	PC1
$\eta(u, \omega, 1)$	29.9%				
$\eta(u, \omega, 2)$	38.2%	39.6%			
$\eta(u, \omega, 3)$	47.0%	46.6%	38.7%		
$\eta(u, \omega, 4)$	62.1%	58.7%	53.7%	40.7%	
$\eta(u, \omega, 5)$	78.3%	71.6%	69.7%	49.3%	44.4%
$\omega_1 = \{PC1, DC2\}$		$\omega_2 = \{DC1, DC3, PC2\}$			
Player u	PC1	DC2	DC1	DC3	PC2
$\eta(u, \omega, 1)$	32.2%		29.1%		
$\eta(u, \omega, 2)$	41.9%	74.5%	31.6%	40.9%	
$\eta(u, \omega, 3)$			58.9%	53.9%	40.8%

Besides, LSP can also adjust the synergy requirement ρ to increase participants' benefits and their willingness to be members of the coalition. The disagreement of several participants on the profit allocation plan would impact negotiations to form a long-lasting collaboration. The uncertainty related to the coalition's stability can also be seen as the influence of participants' endogenous decisions on the robustness of the collaborative network. For instance, the non-adhesion of a distribution or pickup center means that the latter would not contribute in the collective profit, thereby reducing the probability of the coalition to respect SMP principle. Consequently, a suitable synergy requirement can benefit both logistics facilities and coordinators and enable participants to build profitable partnerships.

6. Conclusion

This paper addresses the collaborative multiple centers vehicle routing problem with simultaneous delivery and pickup (CMCVRPSDP). Coalitions are organized by a third-party logistics company which is in charge of conducting negotiations for semitrailer trucks, vehicles and customers sharing. The routing network between DCs, PCs and their respective customers is optimized using HNSGA-II, and CGA method is applied to allocate the collective profits among participants.

The CMCVRPSDP is formulated as a bi-objective mixed-integer mathematical model to minimize cost and the number of vehicles. K-means clustering algorithm and the Non-dominated Sorting Genetic Algorithm-II (NSGA-II) are properly combined into a hybrid heuristic to reduce the network complexity and increase solution quality. The proposed HNSGA-II is designed to accomplish a systematic optimization of networks composed of heterogeneous facilities, i.e. distribution and pickup centers, and to bind the requirements of vehicles returning to pickup centers if there is one pickup unit on their assigned route. Besides, with the incorporation of clustering approach and modifications on the initial population generation, HNSGA-II outperforms the traditional NSGA-II by 5.5% and MOPSO by 11.9% on modified Solomon benchmarks in term of cost solutions. Besides, results provided by algorithm comparisons on a real case study in Chongqing city have also displayed the comprehensive and high performance of HNSGA-II. Therefore, our proposed hybrid heuristic is efficient and relevant in theory and practice. Further, routing optimization results have been considered to study possible profit allocation schemes. CGA has also been compared with MCRS, Shapley and GQP to find out the superiority of CGA due to the close distance between its solution and the core

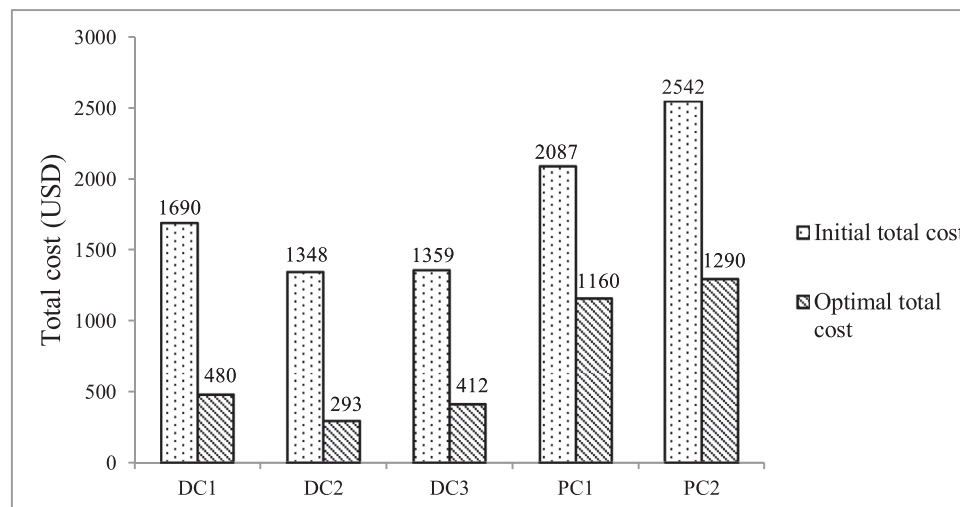


Fig. 8. Grand coalition cost comparison.

center. Finally, using SMP theory, we concluded the efficiency of the formation of a grand coalition instead of two sub-coalitions. After collaboration, the total travelled distance of the entire network, the size of each facility's vehicle fleet, irrational links between suppliers and customers have decreased. Subsequently, logistics companies can not only reach their economic goals, but also maximize service reliability, and ensure a sustainable development.

Future research can consider including the evaluation of the influence of demand uncertainty on the collaborative vehicle routing network as well as its impacts on vehicle sharing. The process of choosing the LSP in the logistics network can also be improved to increase the stability of coalitions. In addition, constraints requiring simultaneous delivery and pickup vehicles to return to pickup centers can be relaxed to make more dynamic decisions. Besides, our approach can be extended to address more precise logistics sectors such as fast consuming goods and cold products supply chains.

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